Distinction Observable Theory. Part III: Physical Realization

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> "The unmanifest [Prakriti, primal matter] is endowed with the three gunas… The gunas, by their nature, are pleasure [Sattva], pain [Rajas], and indifference [Tamas]. Their purpose is to illuminate, to activate, and to restrict. They suppress each other, support each other, produce each other, and act together. Sattva (light) is considered light and illuminating; Rajas (passion) is stimulating and mobile; Tamas (darkness) is heavy and concealing. They work together for a single purpose, like a lamp [where oil, wick, and flame act in concert].
>
> …
>
> Purusha [pure consciousness] is merely a witness to what occurs. It is endowed with absolute isolation, complete indifference, the capacity to contemplate, and absolute non-action.
>
> …
>
> For the sake of Purusha's contemplation of Prakriti, and for its ultimate liberation, their union occurs. It is like the union of the lame and the blind. From this union, creation arises."
> — *Sankhya Karika, 11–21*

> *Tat tvam asi* — "Thou art That." — *Chandogya Upanishad, VI.8.7*

> "And the Word became flesh and dwelt among us, and we beheld His glory, the glory as of the only begotten of the Father, full of grace and truth." — *Gospel of John, 1:14*

Abstract

This third and final volume of the Distinction Observable Theory (DOT; i.e. a theory of observable distinctions) performs the physical readout of the mathematical machinery constructed in Parts I–II. The 64 binary vacuum configurations of the $K(2,2,2)$ octahedron, classified by structural defect $\mathcal{D} \in \{0, 1/3, 1, 4/3\}$, are identified with the complete particle spectrum of the Standard Model: 10 massless states (photon, 8 gluons, vacuum), 36 quark states (6 flavors × 3 colors × 2), 12 lepton doublet states (3 generations × 2 × 2), and 6 heavy gauge bosons (§27). An alphabet of seven prime modes $\{2, 3, 5, 7, 11, 13, 17\}$ is shown to exhaust the physical degrees of freedom, with the fourth generation forbidden by the twin-prime horizon (§28). The lever formula $M = C/\gamma^k \cdot m_e$, parameterized by two integers (C, k), reproduces the masses of leptons, mesons, baryons,

and gauge bosons to accuracies ranging from $10^{-7}\%$ (proton) to $\sim 1\%$ (§30). Quark confinement is derived as a topological theorem: mode 3 (tripletness) is absent from the proton bare base $C = 136 = 2^3 \times 17$ and exists only in its overtone tail — a holographic principle of color charge (§32). The fine-structure constant ($\alpha^{-1} \approx 137.036$, error $< 10^{-6}$), the Weinberg angle ($\sin^2 \theta_W = 2/9$), and the strong coupling ($\alpha_s = 35/297$) are calculated from octahedral geometry (§33). The volume extends DOT to macroscopic physics: time emerges as Hecke-flow of defect transport (§34), gravity as a stationary defect gradient yielding a $1/r$ potential (§35–§40), and cosmological shares ($\Omega_\Lambda = 56/81$, $\Omega_m = 25/81$, $\Omega_b = 4/81$) as topological fractions of the vacuum horizon (§41). No free parameters are introduced beyond the architecture of the carrier itself.

Keywords: particle spectrum, mass formula, quark confinement, fine-structure constant, Hecke flow, defect transport, gravitational constant, cosmological parameters, Standard Model, $K(2,2,2)$ graph.

This is Part III of a three-part series. Part I [arXiv:XXXX.XXXXX] establishes the octahedral carrier and the machine of distinction. Part II [arXiv:XXXX.XXXXX] derives the full mathematical framework from $\partial^2 = 0$.

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§27. Spectral Reality: Physical Reading of the Quantum Vacuum

Part II derived from the single axiom $\partial^2 = 0$:

- the $K(2,2,2)$ octahedron with its Laplacian spectrum $(\lambda_0, \lambda_1, \lambda_2) = (0, 4, 6)$ (§20);
- the projection invariant $\lambda_1/\lambda_2 = 2/3$ (§20.4);
- 64 binary configurations of the vacuum with four defect classes $\mathcal{D} \in \{0, 1/3, 1, 4/3\}$ (§26.1);
- fractional charges as the projection of dichotomy onto the triadic carrier (§25).

Now we ask: **exactly which physical objects inhabit each defect sector?** The answer requires no new mathematics, but a systematic physical reading of the structures already obtained.

27.1. Sector $\boxed{?}$ = 0: Massless States

In §26.1.4, it was established that the sector $\mathcal{D} = 0$ contains exactly 10 configurations. These are states with an ideal balance of axial loads: no single axis is highlighted, inter-axial frustration is absent, and the projection $\Pi_2(x) = 0$ (§20.5).

Let's consider which specific load patterns (n_1, n_2, n_3) realize a zero defect:

Pattern	Configurations	Total weight $w = n_1 + n_2 + n_3$
$(0,0,0)$	1	0
$(1,1,1)$	8	3
$(2,2,2)$	1	6

Eight configurations $(1,1,1)$ represent the choice of one of two poles on each of the three axes: $2^3 = 8$ variants. Algebraically, this is exactly the **spinor representation** of $C\ell_6$: the faces of the octahedron.

[G1] Physical correspondence. The 10 massless configurations are distributed among three physical classes:

Configuration	w	Physical object	Rationale
$(0,0,0)$: total void	0	**Vacuum state**	Not a single active pole; absence of signal
$(1,1,1)$: 8 faces	3	**8 gluons** ($g_1 \ldots g_8$)	Adjoint representation of $SU(3)$: $3^2 - 1 = 8$. Each face is a massless carrier of color interaction
$(2,2,2)$: total saturation	6	**Photon** (γ)	All axes fully activated; full translational symmetry. Scalar singlet $U(1)$

[G1] Nine carriers ($\gamma + 8g$) on ten massless slots leave **one unoccupied** — the vacuum void itself $(0,0,0)$. This state is not a particle; it acts as the zero element of the algebra, the background on which all other signals are registered.

27.2. Sector $\boxed{?}$ = 1/3: Quarks

The sector $\mathcal{D} = 1/3$ is the most numerous: 36 configurations. The physical meaning of the $1/3$ defect: **one axis violates the balance** — its load differs from the other two by exactly one.

[G1] This definition is consistent with the neighbor-boundary reading of the triadic carrier: a fractional electric charge ($\pm 1/3$ or $\pm 2/3$) arises as a projection of dichotomy onto one of the three axes (§25.1), i.e., as an occupied triadic readout with an irreducible hidden debt `1/3`. The remaining two axes are hidden and form the **color degree of freedom** ($\mathbb{Z}/3\mathbb{Z}$, §25.2).

Counting from the physical side:

Component	Value	Explanation
Flavors	6	u, d, s, c, b, t
Colors (per flavor)	3	r, g, b — three elements of the $\mathbb{Z}_3$ remainder
Particle/antiparticle	2	Quark + antiquark
Total	$6 \times 3 \times 2 = 36$	

The coincidence of the number 36 with the number of configurations in the defect sector $\mathcal{D} = 1/3$ is an **exact combinatorial result** of the $K(2, 2, 2)$ graph.

27.2.1. Spectral nature of quark charge

In §25.4, it was shown that quark charges $1/3$ and $2/3$ are projections onto the spectral subspaces V_1 and V_2 at $k = 3$:

- A quark carries a charge $Q = +2/3$ or $Q = -1/3$ when the dichotomous act is projected onto **one** of the three dipole modes (V_1, dim $= 3$). Projection onto one mode out of three gives $1/3$ of the full spectrum.
- An antiquark has an additional charge: $Q = -2/3$ or $Q = +1/3$.

[D] The charge quantization step — $1/3$ — is generated by the dimension of the framework itself ($k = 3$), rather than postulated. Charge is not an independent quantum number; it is **calculated** from the axial structure of the vacuum.

27.2.2. Quark load patterns

The realization $\mathcal{D} = 1/3$ admits four patterns:

Pattern (n_1, n_2, n_3)	Configurations	Total weight
$(0, 0, 1)$ and permutations	6	1
$(0, 1, 1)$ and permutations	12	2
$(1, 1, 2)$ and permutations	12	4
$(1, 2, 2)$ and permutations	6	5

[G1] Four load patterns naturally break the 36 quark states into four **subfamilies**, differing in total weight. Light patterns ($w = 1, 2$) address quarks of the first and second generations (u, d, s, c); heavy ones ($w = 4, 5$) address the third generation (b, t). Transition between subfamilies requires changing at least two axial loads, creating a natural mixing barrier — an analog of the CKM (Cabibbo–Kobayashi–Maskawa) matrix structure.

27.3. Sector ? = 1: Lepton Doublets

The sector $\mathcal{D} = 1$ consists of 12 configurations in which **all three axes carry different loads**. The only admissible pattern:

$$(n_1, n_2, n_3) = (0, 1, 2) \text{ and its permutations}$$

Number of permutations: $3! = 6$ (arranging loads across axes). For each arrangement, the axis with $n = 1$ admits a choice of one of two poles, giving a factor of 2. Total: $6 \times 2 = 12$.

[G1] Physical correspondence. 12 configurations correspond to three lepton doublets:

Component	Value	Explanation
Generations	3	(e, ν_e), (μ, ν_μ), (τ, ν_τ)
Isospin flavor	2	Charged lepton / neutrino
Particle/antiparticle	2	Lepton + antilepton
Total	$3 \times 2 \times 2 = 12$	

A lepton doublet is an object of **full dispersion**: each axis of the Octahedron carries a workload different from the others. This is a maximally individualized state — the lepton sector "sees" all three axes and distinguishes them. This is why neutrinos no longer fit into the background sector $\mathcal{D} = 0$: their oscillations require a non-zero structural defect and naturally assign them to the same sector $\mathcal{D} = 1$ as charged leptons.

[G1] Interpretation of electric charge. In contrast to quarks, whose defect $1/3$ reflects a projection onto one axis, a lepton doublet with defect 1 covers the **entire space** $(0, 1, 2)$ without hidden axes. All three axes are present in the configuration, and the denominator is cleaned (§25.3). The charged component of the doublet gives an **integer** charge ($Q = \pm 1$); the neutrino component gives a neutral electromagnetic readout ($Q = 0$) of the same defect sector. There is no confinement.

27.4. Sector ? = 4/3: Heavy Gauge Bosons

The sector $\mathcal{D} = 4/3$ is minimal: 6 configurations. Possible patterns:

Pattern (n_1, n_2, n_3)	Configurations
$(0, 0, 2)$ and permutations	3
$(0, 2, 2)$ and permutations	3

Common feature: one or two axes are fully saturated ($n = 2$), while the others are empty ($n = 0$). This is a state of **annihilation of choice** — the system has "killed" distinction on some axes, collapsing both poles into one.

[G1] Physical correspondence. 6 configurations are associated with the heavy electroweak sector:

Component	Value	Explanation
Electroweak generators	3	W_1, W_2, W_3 before mixing
Orientations / chiral readings	2	Active and conjugated reading mode
Total	$3 \times 2 = 6$	

After electroweak mixing, this six-dimensional carrier is read as the observed heavy bosons $W^{\pm}$ and Z^0.

[G1] A high defect ($4/3 > 1$) means that the inter-axial stress in these states **exceeds the maximum dispersion** of the lepton sector. Physically, this manifests as the enormous mass of W and Z (on the order of 80–91 GeV), exceeding the masses of leptons and light hadrons by two orders of magnitude.

[G1] The weak interaction is the only one that violates all discrete symmetries (P, C, CP). In the logic of the Octahedron, this is explained by the fact that the configurations $(0, 0, 2)$ and $(0, 2, 2)$ are **radically asymmetrical**: some axes are completely silent, others are completely screaming. Not a single axis has an intermediate load $n = 1$. The lepton "remembers" all three axes; the W-boson knows only two states — absolute silence and absolute scream. This is an algebraic model of parity violation.

27.5. Mass Hierarchy and Spectral Subspaces

Four sectors are linked to the spectral subspaces of the Laplacian (§20.5) hierarchically:

Defect $\mathcal{D}$	Projection $\parallel \Pi_2(x) \overset{2}{\parallel}$	Subspace	Mass register
0	0	Entirely in $V_0 \oplus V_1$	Massless
1/3	small	Weak touch of V_2	Confined (quarks)
1	full	Full V_2 states	Free massive (leptons)
4/3	super-full	Beyond V_2 norm	Superheavy bosons

[G1] This hierarchy is strictly monotonic by $\mathcal{D}$ and coincides with the experimental mass hierarchy:

$$m_\gamma = m_g = 0 < m_q^{\text{curr}} \sim \text{MeV} < m_\ell \sim 0.5 - 1777\text{MeV} < m_{W,Z} \sim 80 - 91\text{GeV}$$

Current quark masses (not constituent) lie in the range $\sim$ 2–$173,000$ MeV, but their fractional defect $1/3$ places them between the massless and lepton sectors in spectral classification. The

heaviness of the t-quark (~ 173 GeV) is a consequence of its deep embedding in the electroweak register (§30), rather than the defect itself.

27.6. Generational Structure

In §25.5, a strong DOT restriction was established: three generations of fermions are generated by three pairs of twin primes $(5, 7)$, $(11, 13)$, $(17, 19)$, while the fourth is forbidden by the horizon $5^2 = 25$.

Now we can specify how the generations are distributed within the defect sectors:

[G2] Lepton generations ($\mathcal{D} = 1$):

Generation	Twin pair	Lepton	Mass (MeV)	Mass ratio
I	$(5, 7)$	e/ν_e	0.511 / m_{ν_e}	1 / neutrino doublet
II	$(11, 13)$	μ/ν_μ	105.658 / m_{ν_μ}	$\approx 207 \approx 56/\gamma$
III	$(17, 19)$	τ/ν_τ	1776.86 / m_{ν_τ}	$\approx 3477 \approx 3456/\gamma^0$

The mass ratios $\mu/e \approx 207$ and $\tau/e \approx 3477$ are not random: they are calculated by the lever formula $M = C/\gamma^k \cdot m_e$ (§26.2) with specific bases ($C = 56$ for μ, $C = 3456$ for τ). The detailed derivation of the bases is the subject of §29; here only the fact is fixed: **generational masses are not postulated, but calculated from integer addresses on the graph**.

[G1] Quark generations ($\mathcal{D} = 1/3$):

Generation	Up quark	Down quark
I	u (2.16 MeV)	d (4.67 MeV)
II	c (1270 MeV)	s (93.4 MeV)
III	t (172,690 MeV)	b (4180 MeV)

The mass hierarchy in the quark sector spans five orders of magnitude ($m_u/m_t \sim 10^{-5}$), with all six masses calculated by the lever formula with individual bases and depths. The enormous spread of masses with the same defect ($1/3$) is a consequence of the difference in bases C and depths k, rather than a difference in defect type.

27.7. Arrow of Time: Orientation of Traversal

[G2] The spectral structure of the Octahedron carries another physical consequence — the directionality of processes.

Borromean prohibitions (§6, Part I) form a closed ring: $D \to F \to C \to D$. This ring is chiral — it has two possible directions of traversal. The choice of direction (orientation) determines the "arrow": physical irreversibility.

Mathematically, orientation is fixed by the sign of the determinant of the axis permutation. The $K(2,2,2)$ octahedron admits two types of faces (two signs of $\varepsilon_1\varepsilon_2\varepsilon_3$), corresponding to two chiralities. A real observer "lives" within one choice of chirality, which breaks T-symmetry.

In the weak sector ($\mathcal{D} = 4/3$), the configurations $(0,0,2)$ and $(0,2,2)$ do not even possess axial symmetry: axis permutation changes the physics. This reflects the experimentally observed violation of CP in weak decays. Full CPT-symmetry is preserved because the octahedron as a whole is invariant under the **full** symmetry $|2O| = 48$ — antipodal inversion compensates for the chiral traversal.

27.8. Summary of §27

§27 has fixed the physical reading of the four defect sectors:

Sector	$\mathcal{D}$	N	Physics	Key property
Massless	0	10	Photon, 8 gluons, vacuum	Zero inter-axial tension
Quarks	1/3	36	6 flavors × 3 colors × 2	Fractional charge, confinement
Lepton doublets	1	12	3 generations × 2 isospin flavors × 2	Free leptons and neutrinos
Heavy bosons	4/3	6	3 electroweak generators × 2 readings	Violation of P, C, CP
Total		**64**	Standard Model	2^6 = full vacuum

Mathematics (§§20–26) gave us the **numbers** $10, 36, 12, 6$ and the defect **values** $0, 1/3, 1, 4/3$. §27 showed that these numbers and values structurally coincide with the zoo of the Standard Model: massless carriers, confined quarks, free lepton doublets, and heavy weak bosons. This correspondence is a combinatorial result of the $K(2,2,2)$ graph, rather than parametric fitting.

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§28. Alphabet of Bases: Simple Axes of the Physical Vacuum

In §23 (Part III), the arithmetic emanation of the natural series from the operator ∂ was proven: prime numbers are indecomposable boundaries, composite ones are forms. In §26.3, each particle

was assigned a morphological vector $\vec{v}(C) = (k_2, k_3, k_5, k_7, k_{11}, k_{13}, k_{17})$. Now we ask: **what is the physical role of each simple axis**, and why is the alphabet cut off at seventeen?

28.1. Seven Primes as Standing Waves of the Octahedron

Each prime number in the factorization of the structural coefficient C or the nodes of the tail denominator N_i (§26.2) is a **basis mode** — a standing wave on the $K(2,2,2)$ graph. The physical meaning of the modes is determined by their role in the spectral structure:

[D] table of physical modes:

Mode	Prime	Physical meaning	Where realized
Dichotomy	2	Spin, basic parity, binary choice	All bases except 27 and 39
Tripletness	3	Color charge, generations, chirality	$432, 72, 162, 48, 39, 27$
Isospin	5	Strong isospin, polarization	$110 = 2 \times 5 \times 11$
Spinor sphere	7	S^7 Hopf, trialities $SO(8)$	$56 = 2^3 \times 7$
Field of choice	11	Pole diversity, field-layer	$110 = 2 \times 5 \times 11$
Confinement anchor	13	Fixation of deep structure	$39 = 3 \times 13$
Closure	17	Eddington number, $136 + 1 = 137 \approx \alpha^{-1}$	$136 = 2^3 \times 17$, $272 = 2^4 \times 17$

Notice that each mode has a concrete algebraic function:

- **Mode 2** (dichotomy) creates the very possibility of distinction — it is the act ∂ acting on each axis. It is present in almost every base because distinction is the primary operation.
- **Mode 3** (tripletness) is responsible for color structure and chirality. Its presence or absence in base C determines whether the quark structure is internal or external (§32).
- **Mode 7** (spinor) realizes the extended Freudenthal space: $\dim(E_7) = 133$, the fundamental representation has dimension $56 = 2^3 \times 7$. Particles with mode 7 in the base (e, μ, u, B^0) possess spinor connectivity — they "remember" the seven-dimensional structure of the internal space.
- **Mode 17** — arithmetic closure: $2^3 \times 17 = 136$, and $136 + 1 = 137 \approx \alpha^{-1}$. This is not a coincidence, but a reflection of the link between the Eddington number of self-interaction ($T_{16} = 16 \times 17/2 = 136$) and the electromagnetic constant.

28.2. Horizon of Primes: Why the Alphabet is Finite

[G1] The alphabet is cut off at the number 17 for two reasons:

Reason 1: twin primes horizon. In §25.5, it was shown that three generations of fermions are generated by pairs of twin primes $(5, 7)$, $(11, 13)$, $(17, 19)$. The next potential pair is $(23, 25)$, but $25 = 5^2$ is composite. The chain breaks: axis 5 has "looped" back. For a new prime ($p = 23$), no dual primitive exists within the 2-horizon, and the fourth generation is not realized. The physical alphabet is limited to the third generation.

Reason 2: parameter space dimension. The Standard Model contains **19 free parameters**. In DOT, each parameter is not a postulate but an address on the graph. Seven primes generate 14 base numbers (§29), each carrying several bits of information. The informational capacity of the alphabet $\{2, 3, 5, 7, 11, 13, 17\}$ is sufficient to address all 19 parameters. Expanding the alphabet to $p = 19$ or $p = 23$ would give redundant degrees of freedom, not filled by observables — empty addresses without particles.

[G2] On the algebro-geometric side: 7 primes define a 7-dimensional integer space of exponents. This space is isomorphic to $\mathbb{Z}^7$, consistent with the 7 extra dimensions of M-theory ($11 - 4 = 7$) and the 7 complex modules of a K3 surface.

28.3. Morphological Passport of a Particle

Each particle is characterized by its **morphological vector** — the set of prime factors in the decomposition of base C:

$$\vec{v}(C) = (k_2, k_3, k_5, k_7, k_{11}, k_{13}, k_{17}), \quad C = \prod_p p^{k_p}$$

[D] Full table of morphological passports:

Particle	C	k_2	k_3	k_5	k_7	k_{11}	k_{13}	k_{17}	k
Electron e	56	3	—	—	**1**	—	—	—	0
Muon μ	56	3	—	—	**1**	—	—	—	1
u-quark	56	3	—	—	**1**	—	—	—	−2
d-quark	432	4	3	—	—	—	—	—	−3
s-quark	4	2	—	—	—	—	—	—	3
c-quark	48	4	1	—	—	—	—	—	3
b-quark	162	1	**4**	—	—	—	—	—	3
t-quark	136	3	—	—	—	—	—	**1**	6
Tau τ	3456	7	3	—	—	—	—	—	0
Pion π^0	72	3	2	—	—	—	—	—	1
Kaon $K^\pm$	72	3	2	—	—	—	—	—	2
Kaon K^0	72	3	2	—	—	—	—	—	2
D^0	272	4	—	—	—	—	—	**1**	2
D^+	272	4	—	—	—	—	—	**1**	2
B^0	56	3	—	**1**	—	—	—	—	4
B_s^0	56	3	—	**1**	—	—	—	—	4
Proton p	136	3	—	—	—	—	—	**1**	2
Lambda Λ	162	1	**4**	—	—	—	—	—	2
Sigma Σ^+	162	1	**4**	—	—	—	—	—	2
Omega Ω^-	18	1	**2**	—	—	—	—	—	4
W-boson	64	**6**	—	—	—	—	—	—	6
Higgs boson	22	~~3~~	**3**	—	—	—	—	—	7

A dash "—" denotes a zero power.

28.4. Clusters: One Form — Different Depths

[D] Leptoquark cluster $C = 56 = 2^3 \times 7$:

Particle	Depth k	Mass (MeV)	Sector
e	0	0.511	Lepton
μ	1	105.658	Lepton
u-quark	−2	2.16	Quark
B^0-meson	4	5279	Meson
B_s^0-meson	4	5367	Meson

Four physically different particles — a lepton, a quark, and two mesons — have **identical** morphological passports $(3, 0, 0, 1, 0, 0, 0)$. The difference between an electron and a B-meson lies not in the structure but in the **depth of observation** (k). What the Standard Model describes as fundamentally different objects, DOT shows as the same topological form read at different registers.

28.5. Grouping of Bases by Families

From the morphological table, four types of bases are distinguished:

[D] Family 1: Purely binary ($C = 2^n$).

C	Particle	Characteristics
4	s-quark ($k = 3$)	Simplest quark
64	W-boson ($k = 6$)	Full vacuum = 2^6

Bases containing only mode 2 carry pure dichotomous information without color, isospin, or closure. The W-boson ($C = 64 = 2^6$) is a special case: its base **equals the number of vacuum states**. It carries no structural information other than distinction itself, explaining its role as a carrier of the **weak** (violating all symmetry) interaction.

[D] Family 2: Binary-triadic ($C = 2^a \times 3^b$).

C	Particle	Characteristics
18	Ω^- ($k = 4$)	Connecting node: 2×3^2
27	H^0 ($k = 7$)	Pure Higgs: 3^3
48	c-quark ($k = 3$)	O_h-group: $2^4 \times 3$
72	$\pi^0, K^\pm, K^0, Z$	Meson-boson base: $2^3 \times 3^2$
162	Λ, Σ^+, b-quark	Chiral base: 2×3^4
432	d-quark ($k = -3$)	Sub-vacuum Ramanujan pixel: $2^4 \times 3^3$

This is the most populated family. The presence of mode 3 means the particle "knows" color structure internally. The Higgs ($C = 27 = 3^3$) is the only particle without mode 2 in this family: it carries pure tripletness without dichotomy, a pure chiral form.

[D] Family 3: Spinor ($C = 2^a \times 7$).

C	Particle	Characteristics
56	e, μ, u, B^0, B_s^0	Freudenthal: $2^3 \times 7$; spinor cluster with sub-vacuum readout for u

The only base with mode 7. The Freudenthal space of dimension 56 is the fundamental representation of E_7, linked to the spinor sphere S^7 and the triality of $SO(8)$.

[D] Family 4: Eddingtonian ($C = 2^a \times 17$).

C	Particle	Characteristics
136	Proton, neutron ($k = 2$)	T_{16}: $2^3 \times 17$
136	t-quark ($k = 6$)	strongest current electroweak readout of the same Eddington base
272	D^0, D^+ ($k = 2$)	Double Eddington (2×136)

Mode 17 links these bases to the Eddington number: $T_{16} = 16 \times 17/2 = 136$ — the total number of pairwise interactions of 16 spinor states. The proton ($C = 136$) addresses the **maximally connected** structure of the vacuum — hence its exceptional stability.

28.6. Missing Modes: Where There is No Isospin and Confinement

[G1] Modes 5, 11, and 13 are **absent** from all direct bases C. They are realized only in the nodes of tail denominators:

- $110 = 2 \times 5 \times 11$ — pole background, containing modes 5 (isospin) and 11 (field of choice). These modes manifest not as the structure of the particle itself, but as its link to the vacuum.
- $39 = 3 \times 13$ — confinement anchor. Mode 13 participates only in the deep layers of the proton tail ($\Omega_2 \sim 48 \times 39$, $\Omega_3 \sim 39^3$). This is the most "hidden" mode — it addresses no particle directly, but fixes the confinement mechanism at the boundary.

28.7. Summary of §28

§28 has established:

1. **Seven prime numbers** $\{2, 3, 5, 7, 11, 13, 17\}$ are the full physical alphabet of spectral modes of the Octahedron.
2. **Morphological vector** $\vec{v}(C)$ — a unique particle passport.
3. **Clusters**: particles with identical passports (e, μ, B^0 — all with $C = 56$) are the same topological form at different depths.

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§29. Physically Realized Bases: Nodes of the Mass Graph

In §23 (Part III), it was derived that the operator ∂ generates the entire natural series. In §26.2, the lever formula $M_{\text{bare}} = C/\gamma^k \cdot m_e$ was introduced, where C is an integer base. From the infinite series of composite numbers, **12 direct particle bases and 3 additional auxiliary tail nodes** are realized in observed microphysics.

29.1. Catalog of Realized Bases

[D] Full list of 12 direct bases and 3 auxiliary nodes:

Nucleus of emanation (direct particle bases):

C	Factorization	Algebraic structure	Addressed particles
4	2^2	Purely binary pivot	s-quark ($k = 3$)
18	$2 \cdot 3^2$	Connecting node	Ω^- ($k = 4$)
27	3^3	Exceptional Jordan algebra $J_3(\mathbb{O})$, link to E_6	H^0 ($k = 7$)
48	$2^4 \cdot 3$	Order of full O_h Octahedron group	c-quark ($k = 3$)
56	$2^3 \cdot 7$	Freudenthal space, $\dim(\mathbf{56}_{E_7})$	e ($k = 0$), μ ($k = 1$), u ($k = -2$), B^0 ($k = 4$), B_s^0 ($k = 4$)
64	2^6	Full binary vacuum $2^6 = 64$	$W^\pm$ ($k = 6$)
72	$2^3 \cdot 3^2$	Meson-boson base	π^0 ($k = 1$), $K^\pm$ ($k = 2$), K^0 ($k = 2$), Z ($k = 6$)
136	$2^3 \cdot 17$	Triangular number T_{16} (Eddington)	p ($k = 2$), n ($k = 2$), strongest current t ($k = 6$)
162	$2 \cdot 3^4$	Chiral base	Λ ($k = 2$), Σ^+ ($k = 2$), b-quark ($k = 3$)
272	$2^4 \cdot 17$	Double Eddington (2×136)	D^0 ($k = 2$), D^+ ($k = 2$)
432	$2^4 \cdot 3^3$	Ramanujan pixel	d-quark ($k = -3$); also vacuum tail-node
3456	$2^7 \cdot 3^3$	Extended Ramanujan base (8×432)	τ ($k = 0$)

Auxiliary nodes (tail denominators):

N	Factorization	Role in tail
24	$2^3 \cdot 3$	Mellin background of light quarks; Leech lattice dim 24
39	$3 \cdot 13$	Confinement anchor; deep layers of proton tail
110	$2 \cdot 5 \cdot 11$	Pole background; only node with modes 5 and 11
432	$2^4 \cdot 3^3$	Ramanujan pixel ($j(i)/4$); denominator of α^{-1} and bleed-target

29.2. Algebra of Relations: Multiplicative Links

The proton ($C = 136$) is the "algebraic center" of the entire system.

§30. Emanation Table: Full Catalog of Bare Masses

In §26.2 (Part III), the lever formula was derived:

$$M_{\text{bare}} = \frac{C}{\gamma^k} \cdot m_e, \qquad \gamma = \frac{\sqrt{6}}{9} \approx 0.27217, \quad m_e = 0.51099895\text{MeV}$$

The bare mass is determined by **two integers** (C, k) and car-ries the main share of the resulting physical mass.

30.1. Lepton Sector

Particle	C	k	M_{bare} (MeV)	PDG (MeV)	Error
$\boldsymbol{e}$	56	0	$56 \cdot m_e = 28.616$	0.51100	[quantum]
$\boldsymbol{\mu}$	56	1	105.14	105.658	**0.49%**
$\boldsymbol{\tau}$	3456	0	1766.0	1776.86	**0.61%**

30.2. Meson Sector

Particle	C	k	M_{bare} (MeV)	PDG (MeV)	Error
$\boldsymbol{\pi^0}$	72	1	135.18	134.977	**0.15%**
$\boldsymbol{K^+}$	72	2	496.68	493.677	**0.61%**
$\boldsymbol{D^0}$	272	2	1876.39	1864.84	**0.62%**
$\boldsymbol{B^0}$	56	4	5215.26	5279.66	**1.22%**

30.3. Baryon Sector

Particle	C	k	M_{bare} (MeV)	PDG (MeV)	Error
$\boldsymbol{p}$	136	2	938.194	938.272	**0.008%**
$\boldsymbol{\Lambda}$	162	2	1117.55	1115.68	**0.16%**
$\boldsymbol{\Sigma^+}$	162	2	1117.55	1189.37	**6.04%**
$\boldsymbol{\Omega^-}$	18	4	1676.33	1672.45	**0.23%**

The dimensionless ratio of the proton bare mass to the electron mass in DOT is the exact integer 1836.

30.4. Boson Sector

Particle	C	k	M_{bare} (MeV)	PDG (MeV)	Error
$\boldsymbol{W^\pm}$	64	6	$80,463.9$	80,360	**0.13%**
$\boldsymbol{Z^0}$	72	6	$90,521.9$	91,188	**0.73%**
$\boldsymbol{H^0}$	27	7	$124,300$	125,250	**0.76%**

30.5. Quark Sector

Quark	C	k	M_{bare} (MeV)	PDG (MeV)	Error
$\boldsymbol{u}$	56	−2	2.12	2.16	**1.9%**
$\boldsymbol{d}$	432	−3	4.45	4.67	**4.7%**
$\boldsymbol{s}$	4	3	101.4	93.4	**8.0%**
$\boldsymbol{c}$	48	3	1217	1270	**4.2%**
$\boldsymbol{b}$	162	3	4107	4180	**1.7%**
$\boldsymbol{t}$	136	6	$170,986$	172,690	**1.0%**

30.6. Registers of Emanation

A key property of the lever formula: **the same base C at different depths k generates physically different particles**. For example, the muon and B-meson share the same spinor address ($C = 56$), read at different scales ($k = 1$ vs $k = 4$).

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§31. Overtone Algebra of Mass

In §26.2.3 (Part III), the concept of an overtone tail was introduced: the bare mass M_{bare} is corrected by a small multiplicative series $1 + \delta$. §30 provided full mass formulas with tails for 16 particles. §31 extracts the **tail algebra** from this data — rules by which numerators, denominators, and signs are constructed.

31.1. General Overtone Tail Formula

[G1] For closed-cases in DOT, the mass is written as a finite sum:

$$\boxed{M = M_{\text{bare}} \times \left(1 + \sum_{n=1}^{N_{\text{cross}}} \frac{((\pm 1)^n)\varphi_n}{\prod_{i=1}^{n} N_i}\right)}$$

This formula is isomorphic to the perturbation theory series in QFT:

DOT Component	QFT Analog
M_{bare}	Tree-level amplitude
φ_n — numerator	Vertex amplitude (coupling constant)
$\prod N_i$ — denominator	Propagator (virtual exchange)
$(\pm 1)^n$ — sign	Fermion loop rule
N_{cross} — number of layers	Loop correction order

[G1] Within canonically closed addresses, the series is observed as **finite** ($N_{\text{cross}} \leq 5$), matches because the $K(2, 2, 2)$ carrier is finite and admits only a finite number of stable routes.

31.1.1. Class Law of Tails

[G1] The tail is not universal in a naive sense. Its form is determined by the **topological isolation class**:

1. **Asymptotic class.** Leptons, mesons, baryons, and weak bosons. Their tails are built from full vacuum nodes $\{432, 136, 110, 39\}$ and admit a multi-layered structure.
2. **Confinement class.** Quarks. Their tails do not exit onto a full free vacuum route and use a short internal grid $\{12, 24, 56\}$.
3. **Electroweak boundary class.** Top quark and heavy bosons ($k = 6$).

31.1.2. Two-Layer Law of the Confinement Tail

For the five confined quarks, the current best tail form is:

$$\delta_q = \delta_{\text{grid}} + \Omega_{\text{bleed}} = \frac{\sigma_q}{N_q} + \frac{\rho_q}{T_q}, \qquad q \in \{u, d, s, c, b\}.$$

The five confined quarks form a nearly symmetrical **grid-layer** pattern:

$$+\frac{1}{56}, \quad +\frac{1}{24}, \quad -\frac{1}{12}, \quad +\frac{1}{24}, \quad +\frac{1}{56}.$$

31.2. Number of Layers: N_{cross}

[D] The number of overtone layers is determined by the interlacing rule:

$$N_{\text{cross}} = \min\big(\ k, \#\text{distinct_primes}(C)\big)$$

N_{cross}	Particles
1	μ, τ, π^0, W, H^0
2	$K^\pm, K^0, D^0, D^+, Z, B^0, B_s^0, \Lambda, \Sigma^+, \Omega^-$
5	p (proton)

31.3. Mellin Numerators (Phase Angles φ_n)

Tail numerators fall into two classes:

Discrete class (rational fractions)

φ	Physical meaning
$2/3$	Electric charge; Koide invariant $9\gamma^2$
$1/3$	Triplet quark asymmetry
$1/2$	Spin-1/2, fermion shift
$1/4$	Orthogonal shift ($\pi/2$)
1	Full presence

The $2/3$ invariant dominates: it appears in the tails of 9 out of 16 key particles.

Continuous class (irrational)

φ	Physical meaning
π	Full cyclic traversal
$\sqrt{6}$	Octahedron edge length
γ	Scale invariant $\sqrt{6}/9$
γ^2	Invariant square $2/27$
α^2	2nd order EM interaction quantum
$R_{av}\gamma = 2\gamma/3$	Average field on the Octahedron

31.4. Denominators: Propagator-Routes

[D] Denominators are the DOT graph nodes:

Node N	Origin
432	Ramanujan pixel: $2^4 \cdot 3^3$
136	Nucleon base: $2^3 \cdot 17$ (quantum $137 - 1$)
110	Frustration cross-section
72	Meson octave: $2^3 \cdot 3^2$
56	Spinor base: 8×7
48	Order of full symmetry group $\parallel 2O \parallel$
39	Confinement anchor 3×13

31.5. Alternation of Discrete and Continuous (D ↔ C)

In key particles, a strict alternation of numerator types (D = discrete, C = continuous) is observed:

Particle	Layer 1	Layer 2	Layer 3	Layer 4
D^0	D: $2/3$	C: $2\gamma^2/3$	D: $1/4$	—
p	D: R_{pole}	C: $R_{av}\gamma/4$	D: 4	C: π
Z	D: 1	C: $\alpha^2/2$	D: $1/3$	—
W	C: $R_{av}\gamma$	D: $2/3$	C: $\pi/2$	—

31.6. Bidirectionality: Mass Standing Wave

Half of the particles exhibit a sign change between tail layers. The particle undergoes **decaying oscillations** around its true mass. Each overtone is a "swing" in one direction or the other,

weakening geometrically (by a factor of N at each step). This is a discrete-spectral analog of a standing wave at the "particle–vacuum" boundary.

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§32. Confinement: Arithmetic of Color Charge

In §25 (Part III), it was established that fractional charges $1/3$ and $2/3$ are dichotomous projections onto the triadic carrier. In §27.2, it was shown that the defect sector $\Delta = 1/3$ contains exactly 36 quark states. Now we ask: **why are quarks not observed freely?** The answer lies in the arithmetic of bases — in the presence or absence of **mode 3** in the factorization of the structural coefficient.

32.1. Arithmetic of the Proton

[D] Proton bare base: $C = 136 = 2^3 \times 17$.

Factorization contains only modes 2 (dichotomy) and 17 (closure). **Mode 3 (tripletness) is absent.** The internal informational structure of the proton is "blind" to the quark trio.

However, in the overtone tail, mode 3 is present powerfully. In base $C = 136$: 3^0 (null). In the tail: 3^{14}. The entire quark content of the proton is **entirely in the overtone structure**, on the particle–vacuum boundary.

32.2. Holographic Principle of Confinement

[G1] From the arithmetic in §32.1, a model isomorphic to the holographic principle (AdS/CFT) follows:

Layer	Space	Content	Mode 3
Bulk	Bare base $C = 136$	Stable, "colorless" structure	❌ Absent
Boundary	Overtone tail	Quark structure, color charge	✅ Powerful

Attempting to "tear" a quark from a proton is equivalent to trying to tear a two-dimensional shadow from a three-dimensional object. Quarks do not exist inside the proton, but **on its phase boundary with the vacuum** — as a holographic projection of the bulk structure onto the surface.

32.3. Two Types of Baryons

Baryons fall into two topological types according to the factorization of base C:

Type I: External tripletness (p, n, $C = 136$). Mode 3 is absent in the base, appearing only in the tail. Quark structure is a holographic projection. Result: **maximum stability**.

Type II: Internal tripletness (Λ, Σ, Ω, $C = 162$ or 18). Mode 3 is present in the base (3^4 or 3^2). The quark trio is part of the internal structure. Result: **instability** ($\tau \sim 10^{-10}$ s).

32.4. Stability of the Proton as a Topological Theorem

[G1] Proton stability in DOT is not an empirical fact, but a theorem: 1. The proton base $C = 136 = 2^3 \times 17$ does not contain mode 3. 2. All quark characteristics are properties of the tail, not the bulk. 3. For the proton to decay, the base C must change (transition to another alphabet node). 4. The nearest nodes $(72, 64, 162)$ lie at energy barriers $\Delta C \geq 64$, excluding decay.

32.5. Weinberg Angle

[D] From the Laplacian spectrum of the Octahedron (§20), the bare value of the electroweak mixing angle is calculated:

$$\sin^2 \theta_W = \frac{2}{9} = 3\gamma^2 \approx 0.22222$$

The formula $2/9 = 3\gamma^2$ links electroweak mixing to the geometry of the Octahedron: the octahedron and its dual cube possess exactly **9 planes of symmetry**. In DOT-reading, 2 of them are read as electromagnetic, while the remaining 7 are read as weak.

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§33. Spectral Audit

Sections §27–§32 have established the full apparatus of physical emanation: from spectral sectors (§27) through the base alphabet (§28–§29), the mass catalog (§30), overtone algebra (§31), and the confinement mechanism (§32). §33 summarizes: **how accurately does DOT calculate observed values**, and what questions remain open.

33.1. Summary Table of Mass Accuracy

Particle	Class	Layers	Δ (MeV)	Error (%)	n_σ	Status
$\boldsymbol{p}$	Baryon	**5**	$+0.000,000,4$	0.000,000,4%	**1.4σ**	✅
$\boldsymbol{\mu}$	Lepton	2	$-0.000,006$	0.000,006%	**2.66σ**	✅
$\boldsymbol{\pi^0}$	Meson	3	$-0.000,24$	0.000,18%	**0.48σ**	✅
$\boldsymbol{B^0}$	Meson	3	$-0.000,49$	0.000,009%	**0.004σ**	✅
$\boldsymbol{W}$	Boson	3	$+0.41$	0.000,51%	**0.03σ**	✅
$\boldsymbol{Z}$	Boson	3	$+0.0169$	0.000,0186%	**0.01σ**	✅

Confinement Grid: Quarks

Quark	M_{cand}	PDG	Error	Status
u	2.1600 MeV	2.16	$\sim 0.00021\%$	✅
d	4.6684 MeV	4.67	$\sim 0.03\%$	✅
s	93.42 MeV	93.4	$\sim 0.0076\%$	✅
c	1270.33 MeV	1270	$\sim 0.0117\%$	✅
b	4180.08 MeV	4180	$\sim 0.0124\%$	✅

33.2. Derived Micro-constants

Parameter	DOT Formula	DOT Value	Experiment	Error
α^{-1}	$\frac{432}{\pi} - \frac{47\gamma}{27}$	137.0361	137.0360	**0.00007%**
$\sin^2\theta_W$ (bare)	$\frac{2}{9} = 3\gamma^2$	0.22222	0.22320	0.44%
Koide K_4	$\frac{2}{3} = 9\gamma^2$	0.66667	0.66667	**0.000%**
$\alpha_s(M_Z)$	$35/297$	0.11785	0.11790	0.05%

All five constants are derived **algebraically or as structural correspondences** from the geometry of the Octahedron and the j-invariant of the elliptic curve.

33.3. Summary of Part III

Part III (§§27–33) has read the mathematical apparatus of Part II as a physical spectrum. All results were obtained from **a single initial object** — the $K(2,2,2)$ graph — through arithmetic emanation and the overtone tail. Not a single parameter was postulated from outside.

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§34. Hecke-Flow: Time as Defect Transfer

In §22 (Part III), the modular flow on the upper half-plane $\mathbb{H}$ was introduced. Now we ask: **what is the physical meaning of this movement?**

[G1] On the $K(2,2,2)$ carrier, each defect ($\mathcal{D} > 0$) is a localized tension — a source of "pressure" in the distinction field. This tension cannot remain stationary: the operator ∂ (Hecke operator T_p) acts as a **propulsion pump**, redistributing the defect across the graph.

[G1] High-frequency micro-pulses of the Hecke operator (T_p) are read by a macroscopic observer as the **flow of time**.

1. **Micro-level**: A discrete sequence of arithmetic operations (multiplication by p, shift by b).
2. **Macro-level**: A smooth, one-way parameter t.

The "arrow of time" is fixed by the irreversibility of the defect transfer from the source (the particle) to the infinite sink (the vacuum). Time does not exist "in itself" as a background; it is a **dynamic readout** of the modular evolution of the defect.

34.4. Gravity as a Defect Gradient

[G1] Gravity in DOT is understood as a **stationary defect-gradient**. The defect density profile created by a mass generates an attractive $1/r$ potential.

Thus, Part V does not introduce time and gravity from the outside, but derives them from the same DOT "machine":

$$\text{micro defect} \to \text{Hecke transport} \to \text{shell dilution} \to \text{Newton-like macro law}$$

---

§35. Effective Defect Transport Equation

§34 fixed three facts: local mass in DOT is a defect, macro-time is a sequence of Hecke-ticks, and shell-current conservation gives a $1/r^2$ dilution law. §35 introduces the **first macro-transport equation**, describing the evolution of defect density.

35.1. Discrete and Continuous Transport

In discrete Hecke-time, the shell-density satisfies a difference transport law:

$$\boxed{\frac{\rho_{n+1} - \rho_n}{\Delta t_n} = -\Gamma_n \rho_n}$$

where $\Delta t_n = \log p_n$ is the Hecke time-step.

In the effective 3D carrier, the natural continuous notation for the transport law is the **continuity equation** for the radial flux:

$$\boxed{\partial_t \rho_D + \frac{1}{r^2} \partial_r (r^2 v_H \rho_D) = 0}$$

where v_H is the effective radial transport velocity induced by the Hecke-flow.

35.2. Stationary Regime and $1/r^2$ Profile

For a stationary macro-profile ($\partial_t \rho_D = 0$), the continuity equation implies:

$$r^2 v_H(r) \rho_D(r) = Q_D$$

where Q_D is the conserved defect flux of the source.

Hence, the defect density:

$$\boxed{\rho_D(r) = \frac{Q_D}{r^2 v_H(r)} \sim \frac{1}{r^2}}$$

This confirms that Newtonian asymptotics are a stationary solution to the transport equation.

35.3. Potential and the Macro-Transport Constant

The potential $\Phi(r)$ is defined as the integral of the defect density:

$$\Phi(r) \sim -\frac{\mathcal{G}_H Q_D}{r}$$

where a candidate for the DOT macro-transport constant is introduced:

$$\boxed{\mathcal{G}_H := \frac{\chi_D}{v_\infty}}$$

Here χ_D is a coupling coefficient, and v_∞ is the asymptotic transport velocity. Identifying $\mathcal{G}_H$ with the Newtonian G requires fixing the normalization of the charge Q_D and the dimension of χ_D.

35.4. Summary of §35

DOT obtains not only static microphysics but also an internal macro-dynamic equation:

$$\text{local defect} \to \text{Hecke transport} \to \text{continuous transport equation} \to \text{stationary } \ 1/r^2 \ \text{ profile}$$

---

§36. Universalization of the Transport Constant

§35 gave the first continuous form of macro-dynamics. To turn the transport coefficient $\mathcal{G}_H = \chi_D / v_\infty$ into a universal macro-constant (analog of G), three conditions must be met.

36.1. Defect Source Charge and Normalization

In DOT microphysics, mass is a defect sector. At the macro level, a **defect source charge** Q_D is introduced — a conserved radial defect flux:

$$Q_D = r^2 v_H(r) \rho_D(r)$$

A linear relationship between the macro-charge and the micro-mass (mass-readout) is assumed:

$$Q_D \propto M_{\text{bare}}^{\text{DOT}}$$

36.2. Universality of Asymptotic Velocity

The key hypothesis (§36.3) states that at large scales, the Hecke-flow transitions into a universal isotropic regime, where the **asymptotic transport velocity** v_∞ does not depend on the source type.

This turns $\mathcal{G}_H$ from a class-specific coefficient into a candidate for a universal macro-constant.

36.3. Candidate Macro Constant

When the universality conditions are met, the effective gravitational constant in DOT takes the form:

$$\boxed{G_{\text{eff}} := \eta_M \mathcal{G}_H = \eta_M \frac{\chi_D}{v_\infty}}$$

where η_M is the source normalization coefficient.

The potential assumes a universal form:

$$\boxed{\Phi(r) \sim -G_{\text{eff}} \frac{M}{r}}$$

36.4. Summary of §36

§36 completes the minimal macro-dynamic framework. Gravity enters the DOT canon for the first time not as an external postulate, but as a **direct consequence of defect transport**. The theory now has a structural "slot" for a macroscopic coupling constant.

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§37. Extraction Problem: Toward the Macro Constant

§36 established the structural "slot" for the macro-constant. §37 formulates the task of its numerical extraction from the internal geometric and operator layers of DOT.

37.1. Four Rigid Elements

From §§34–36, it follows: 1. **Carrier-level transport**: Hecke coefficients λ_p and conductivity κ_p. 2. **Stationary far-field law**: $1/r^2$ dilution law and $-1/r$ potential. 3. **Source normalization**: link between micro-mass and macro-defect charge ($Q_D = \eta_M M$). 4. **Candidate macro constant**: $G_{\text{eff}} = \eta_M \frac{\chi_D}{v_\infty}$.

The task reduces to determining three values: η_M, χ_D, v_∞.

37.2. Three Sources of Normalization

1. **Geometric carrier-scale**: provides dimensionless invariants (e.g., γ).

2. **Hecke transport scale**: determines the transport velocity v_∞ (the denominator of the macro-constant).
3. **Micro-to-macro source bridge**: determines the "strength" of the local mass entry into the field (η_M, χ_D).

The macro-constant G_{eff} in DOT is the **ratio of source normalization to the universal transport scale**.

37.3. Canonical Extraction Criterion

A formula for G_{eff} is considered canonically extracted only if it: - uses only internal DOT objects; - introduces no new free constants; - is invariant under Hecke averaging; - yields a law of gravitation of the form $\Phi(r) \propto -M/r$.

37.4. Summary of §37

The problem of the gravitational constant in DOT has been moved from the realm of philosophical questions to the realm of a **normalization task**:

$$\boxed{G_{\text{eff}} = \eta_M \frac{\chi_D}{v_\infty}}$$

The task of the next stage is to extract these values from the modular spectrum.

---

§38. Minimal Extraction Attempt for the Macro Constant

§37 set the task of extracting the macro-transport constant G_{eff} from the introduced DOT invariants. §38 makes the first canonical attempt at such an assembly, using only carrier objects, spectral Hecke coefficients, and source normalization.

38.1. Reduction to Two Unknowns

The following simplified structure is proposed for the macro-constant:

$$\boxed{G_{\text{eff}} = \frac{\Xi_D}{v_\infty}}$$

where $\Xi_D = \eta_M \chi_D$ is the aggregate "source–field" coupling coefficient, and v_∞ is the asymptotic Hecke transport velocity.

38.2. Candidate Invariants

1. **Denominator (v_∞)**: The best candidate is the averaged isotropic Hecke conductivity:

$$\boxed{v_\infty = \langle \kappa \rangle_{\text{iso}}}$$

where $\langle\kappa\rangle$ is the mean value of the coefficients κ_p in the limit of large primes.

2. **Numerator (Ξ_D)**: It is assumed that the coupling is given through the universal geometric invariant of the Octahedron γ:

$$\boxed{\Xi_D \propto \gamma^m}$$

38.3. Minimal Closed-Form Template

Combining the elements, we obtain a **canonical formula template**:

$$\boxed{G_{\text{eff}} \sim \frac{\gamma^m}{\langle\kappa\rangle_{\text{iso}}}}$$

Here the numerator must come from the carrier geometry, and the denominator from its asymptotic transport.

38.4. Summary of §38

DOT now has not just an idea, but a **rigorous extraction scheme**:

$$\text{carrier}\to\gamma \qquad \text{Hecke transport}\to\langle\kappa\rangle_{\text{iso}} \qquad \Rightarrow \qquad G_{\text{eff}} \sim \frac{\gamma^m}{\langle\kappa\rangle_{\text{iso}}}$$

The task of the next stage (§39) is to verify whether the exponent m and the mean $\langle\kappa\rangle$ can be calculated without external calibration.

---

§39. Asymptotic Kappa and the Dimensionless Macro Skeleton

§38 established the template for the canonical macro-constant formula. In §39, an attempt is made to calculate its dimensionless "skeleton" based on the isotropic averaging of Hecke coefficients.

39.1. Isotropic Hecke Conductivity

To estimate $\langle\kappa\rangle_{\text{iso}}$, the distribution of normalized eigenvalues of the Hecke operator $\lambda_p = 2\cos\theta_p$ is used. The asymptotic isotropic mean is:

$$\boxed{\langle\kappa\rangle_{\text{iso}}^{\text{mod}} = \frac{4}{3\pi} \approx 0.424413}$$

This value is the strongest candidate for the universal conductivity of macro-transport.

39.2. Exponent m

Since the defect source in DOT is quadratic ($D \sim \| \Pi_2 \|^2$), and the universal scale invariant of the carrier is γ, the most natural choice is:

$$\boxed{m = 2}$$

39.3. Dimensionless Macro Skeleton

Combining the carrier scale ($\gamma^2 = 2/27$) and the transport conductivity ($4/3\pi$), we obtain the **dimensionless skeleton of the macro-constant**:

$$G_{\text{eff}}^{(0)} \sim \frac{\gamma^2}{\langle \kappa \rangle_{\text{iso}}} \sim \frac{2/27}{4/(3\pi)} = \frac{\pi}{18}$$

Numerically:

$$\boxed{G_{\text{eff}}^{(0)} \sim \frac{\pi}{18} \approx 0.174533}$$

39.4. Summary of §39

DOT has received its first calculable form of a macro-constant. This is not the final physical constant in SI units, but its internal "skeleton," linking the geometry of the octahedron with asymptotic modular transport.

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§40. Candidate Dimensional Bridge

§39 fixed the dimensionless macroscopic skeleton $G_{\text{eff}}^{(0)} \sim \pi/18$. §40 makes an attempt to link it with the observed dimensionless gravitational coupling constant α_G.

40.1. Gravitational Coupling in Electron Units

The goal is the dimensionless constant:

$$\alpha_G := \frac{G m_e^2}{\hbar c} \approx 1.75181 \times 10^{-45}$$

In DOT, the scale 10^{-45} arises as a consequence of **horizon depth** and binary weakening.

40.2. Horizon Depth and Binary Link

The strongest candidate for the macroscopic horizon in DOT is the node **39** ($k_\Omega = 39$). The gravitational link connects a "source–detector" pair, giving a binary weakening factor:

$$\eta_M = (\gamma^{k_\Omega})^2 = \gamma^{78}$$

40.3. Refined Reference Formula

Combining the macro-skeleton ($\pi/18$), the arithmetic intertwining factor ($12/\pi^2$), and the binary weakening (γ^{78}), we obtain a refined calculation formula:

$$\alpha_G^{\mathrm{DOT,ref}} = \frac{2}{3\pi}\gamma^{78} = \frac{2^{40}}{\pi \cdot 3^{118}} \approx 1.75284 \times 10^{-45}$$

The deviation from CODATA experimental data is only **0.06%**.

40.4. Summary of §40

The scale of gravity (10^{-45}) in DOT is not postulated but derived from the carrier architecture (depth $k = 39$) and binary defect transport. This completes the construction of the "bridge" from microphysics to macroscopic gravitation.

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§41. Macroscopic Horizon Shares

DOT reads the Universe through pure topological shares of the full vacuum denominator of the horizon. At this level, the cosmological balance decomposes into stable fractions without invoking external entities.

41.1. Vacuum Denominator of the Horizon: 81

The DOT macroscopic horizon is read on the denominator **81** (3^4), which is an enhanced triadic depth. The total sum is normalized as $\Omega_{\mathrm{tot}} = 81/81 = 1$.

41.2. Canonical Macro-Shares (Ω)

The main parameters of the DOT cosmological model are given by the following shares: - **Dark energy**: $\Omega_\Lambda = 56/81 \approx 0.691$ - **Total matter**: $\Omega_m = 25/81 \approx 0.309$ - **Baryon matter**: $\Omega_b = 4/81 \approx 0.049$ - **Dark matter**: $\Omega_c = 21/81 \approx 0.259$

These shares close perfectly: $56 + 25 = 81$ and $4 + 21 = 25$.

41.3. Comparison with Experimental Data

Parameter	DOT Formula	DOT Value	Planck/CODATA
Ω_Λ	$56/81$	0.6914	0.685 ± 0.007
Ω_m	$25/81$	0.3086	0.315 ± 0.007
Ω_b	$4/81$	0.0494	0.0493 ± 0.0008
Ω_c	$21/81$	0.2593	0.2642 ± 0.0045

For the baryon share, the relationship $\Omega_b = (2/9)^2$ holds, which agrees with the electroweak sector of DOT.

41.4. Summary of §41

The Universe in DOT is a single machine of distinction, where local registers ($k = 0...7$) yield microphysics, and the deep horizon ($k = 39, 3^4 = 81$) yields macroscopic shares. Neither requires external parameters outside the carrier architecture.

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§42. Algebraic Closures

§42 places an algebraic foundation under the physical and cosmological parameters of DOT, highlighting strict mathematical cores (status [D]) from their physical interpretations ([G1]).

42.1. Theorem on the Number 47

The number **47**, which enters the formula for the fine-structure constant, is strictly fixed as the dimension of the non-trivial complement to the vacuum subspace in the octahedron symmetry group O_h:

$$\dim(\mathbb{C}[O_h] \ominus V_{\text{triv}}) = 48 - 1 = 47$$

This provides an algebraic support for the correction term α_{DOT}^{-1}.

42.2. Closing the Macro-Horizon: 81

The horizon denominator **81** admits a decomposition in the external algebras of the spinor and observable spaces:

$$\dim \Lambda^3(\mathbb{R}^8) + \dim \Lambda^2(\mathbb{R}^7) + \dim \Lambda^3(\mathbb{R}^4) = 56 + 21 + 4 = 81$$

This confirms the structure of cosmological shares $\Omega_\Lambda = 56/81, \Omega_c = 21/81$ and $\Omega_b = 4/81$.

42.3. Combinatorics of the Mediator Graph: 78 and 39

If the macroscopic vacuum is described by a complete mediator graph K_{13}, its parameters are fixed rigidly: - Number of binary links: $E(K_{13}) = \binom{13}{2} = 78$ - Single-sided readout depth: $78/2 = 39$ These numbers set the weakening factor γ^{78} and the horizon depth $k_\Omega = 39$, used in the gravitational bridge (§40).

42.4. Summary of §42

Algebraic closures make the theory significantly more rigid. DOT does not simply "fit" numbers to experiment, but shows their deep rootedness in the symmetries and combinatorics of the carrier.

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